

# Pankaj Singh

Ph.D. Mathematics • Formal Methods • Lean 4 Formalization • AI for Mathematics

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## SUMMARY

Mathematics Ph.D. (defense June 2026) with research-level training in algebraic and arithmetic geometry, logic-intensive teaching experience, and hands-on experience formalizing mathematics in Lean 4 with Mathlib. Recent work spans a four-person Lean 4 formalization of the Lorenz-system trapping-region lemma (Lyapunov methods, invariant regions, and reusable topology lemmas targeting Mathlib) and an audit methodology for AI-generated Lean formalizations: statement fidelity, assumption hygiene, proof faithfulness, and semantic drift, directly addressing the reliability of human–AI collaboration in proof-assistant libraries. Strongly motivated by FORM’s vision of a universal, proof-assistant-verified repository of computing knowledge, and eager to contribute research-level formalization, library design, and evaluation methodology to efforts such as CSLib.

## FORMAL METHODS, LEAN FORMALIZATION & VERIFIABLE AI

### Lorenz-System Trapping Region in Lean 4 Team formalization (group of four)

2026

ICARM Summer School on Formalization of Mathematics

[icarm.io/summer-school-2026](https://icarm.io/summer-school-2026)

- Formalized, with a four-person team, the classical trapping-region argument for the Lorenz system in Lean 4 / Mathlib: the vector field and solution predicate, the Lyapunov function  $V$  and its derivative  $\dot{V}$  along trajectories, compactness of the region  $\{\dot{V} \geq 0\}$ , and forward invariance of the sublevel trapping region  $\{V \leq \max_D V + 1\}$ .
- Co-authored a set of reusable, domain-agnostic *Mathlib-candidate* lemmas on preconnected sets, frontiers, and continuous first-exit times e.g. a continuous image of a preconnected set meeting both a set and its complement must meet the set’s frontier, and the first-exit point of a continuous path lies on the frontier the topological engine driving the invariance proof.
- Established the analytic core *sorry-free*:  $C^\infty$  smoothness of  $V$  and  $\dot{V}$ , the exact algebraic identity for  $\dot{V}$  along Lorenz solutions, and closedness, boundedness, and compactness of the sign region  $D$  and trapping region via Heine–Borel.

### Kontsevich Deformation Quantization in Lean Core Researcher (ongoing)

2026–present

Berkeley Research project group (Track 4); mentors: Erik Herrera and Bogdan Dina

- Contributing to a research-level Lean 4 formalization stream involving graph-based and representation-theoretic configurations, with connections to Mackey-functor methods, torsors of tori, and autoformalization.
- Gaining sustained experience in collaborative, library-scale formalization: API design, Mathlib conventions, and structuring definitions for reuse by other contributors.

### Improving Automated Theorem Proving Hummingbird Bioscience

2026

Math to Power Industry (PIMS / M2PI), Team Autoformalization

[m2pi.ca/project/2026/hummingbird](https://m2pi.ca/project/2026/hummingbird)

- Audited AI-generated Lean 4 formalizations to distinguish clean compilation from genuine preservation of mathematical meaning, developing a reusable failure taxonomy and audit workflow.
- Evaluated formal proof attempts along three core audit categories: **statement fidelity**, **assumption hygiene**, and **proof faithfulness**, including axiom/*sorry* dependency checks.
- Worked with LLM-assisted Lean workflows: Lean 4, Mathlib, Lean Language Server, MCP-style proof-state feedback, and AI coding systems (Claude, DeepSeek, Codex-style agents).
- Identified assumption-migration failures where hard mathematical content was displaced into hypotheses, definitions, interfaces, comments, or overgeneralized helper statements.
- Analyzed a research-level case study on flasque and coflasque resolutions of  $G$ -lattices, correcting a false universal  $H^1$ -vanishing statement and documenting remaining proof obligations.
- Proposed evaluation tooling for trustworthy autoformalization: theorem-statement alignment, assumption-provenance graphs, semantic-definition drift detection, proof-strategy preservation tests, and source-to-code audit trails.

## TECHNICAL SKILLS

**Proof assistants & formal methods:** Lean 4, Mathlib, Lean Language Server, MCP-based agent tooling, formal proof auditing, proof-assistant evaluation workflows

**Logic & foundations:** first-order logic, proof theory in practice (taught logic and introduction to proofs), type-theoretic formalization in Lean

**Programming & computation:** Python, SageMath, Macaulay2, Git/GitHub, L<sup>A</sup>T<sub>E</sub>X

**AI / ML:** LLM-assisted formalization workflows, agentic AI coding, applied ML (regression, classification, gradient descent, overfitting mitigation)

**Mathematical domains:** algebraic & arithmetic geometry, commutative algebra, group cohomology, Brauer groups, algebraic tori, torsors, rationality problems, dynamical systems / ODE flows

## MATHEMATICAL RESEARCH

- **Classifying Torsors of Tori with Brauer Groups**  
with Alexander Duncan. *Journal of Pure and Applied Algebra* **230**(9), Article 108325, 2026.
- **On the Weak Lefschetz Property for Ideals Generated by Powers of General Linear Forms**  
with Matthew Booth and Adela Vraciu. To appear, *Journal of Commutative Algebra*. [arXiv:2410.22542](https://arxiv.org/abs/2410.22542)
- **Torsors of Generalized del Pezzo Tori and Brauer Groups (submitted)**  
Preprint. [arXiv:2606.16065](https://arxiv.org/abs/2606.16065)

## EDUCATION

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|-----------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>Ph.D. in Mathematics</b> , University of South Carolina<br><i>Advisor: Dr. Alexander Duncan</i>                                | Aug 2020 – Jun 2026 |
| <b>M.S. in Mathematics</b> , Central Michigan University<br><i>Thesis: Connectivity of the Tropical Double Ramification Cycle</i> | Aug 2018 – Jun 2020 |
| <b>M.Sc. in Mathematics &amp; Computing</b> , IIT Guwahati                                                                        | Aug 2014 – Jul 2016 |

## TEACHING

Instructor of record for 14 university courses (2019–2026), including **MATH 174: Discrete Mathematics for Computer Science** (logic and introduction to proofs), calculus, and algebra courses of up to 60 students; experienced in mentoring new graduate instructors and in curriculum development with generative AI.

## SELECTED TALKS, SERVICE & FIELD BUILDING

- **Selected talks:** AI  $\times$  Mathematics 2026, ICMS Edinburgh; Short Communication accepted, International Congress of Mathematicians, Philadelphia (2026); Joint Mathematics Meetings, Washington, D.C. (2026); invited talk, AWM Research Symposium, UW–Madison (2025).
- **Professional development:** Participant, Summer School on Formalization of Mathematics, Institute for Computer-Aided Reasoning in Mathematics (ICARM), July 2026; Reimagining Assessments with Generative AI (2024).
- **Organizer:** AMS Fall Eastern Sectional Special Session “Number Theory Meets Algebraic Geometry” (2025); Reading Seminar on Toric Geometry in Mirror Symmetry (2025); Graduate Colloquium and AG/CA/NT Student Seminar, University of South Carolina.
- **Reviewer / Judge:** Undergraduate Poster Presentations at Joint Mathematics Meetings (2026); USC Region II Science & Engineering Fair (2025).